A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 03

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Unburnt fuel in vehicle exhaust gases **1** Evaporation from fuel tanks / spillage at filling stations **1** Solvents (paints, cleaning products) **1** Aerosol propellants **1** Gas leaks / fossil fuel exploitation (methane) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank **1** Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe **1** Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) **1** NOx are chemically reduced to nitrogen and oxygen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues **1** Brain damage / death if enough haemoglobin is inactivated – low doses worsen chronic heart disease **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** $3.6 \times 10^{11} / 1.2 \times 10^{9}$ **1** = 300 years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** It must be porous, with spaces between the particles, so that it can store a large volume of water **1** It must be permeable, so that water can flow through it to a well or spring fast enough to be abstracted **1** An impermeable layer beneath it stops the water draining away, and the rock must be exposed at the surface somewhere, or overlain by permeable material, so that rainfall can recharge it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **permeable** → how easily water passes through / lets water through • **impermeable** → does not let water through	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Abstraction of 4.8×10^{9} m^3 per year is four times the recharge of 1.2×10^{9} m^3 per year, so water is being removed faster than it is replaced **1** The stored water is being mined rather than used sustainably, so the saturated zone shrinks and its upper surface, the water table, falls each year **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Abstraction** → water taken out of a river, lake or aquifer	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Wells and boreholes must be deepened and the water pumped from further down, so the energy cost of abstraction rises and shallow wells fail first **1** Rivers, springs and wetlands fed by the aquifer dry up, destroying habitats and the species that depend on them **1** Near the coast, sea water is drawn into the aquifer (saltwater intrusion), making the water too saline for drinking or irrigation **1** The rock compacts as the water that supported it is removed, causing land subsidence that damages buildings and drainage, and permanently reduces the storage capacity **1** Irrigated agriculture that depends on the aquifer fails, with loss of food production and livelihoods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
02.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Fossil water fell as rain thousands of years ago under a wetter climate – under the present climate recharge is negligible, so any abstraction depletes a finite store and the resource is effectively non-renewable **1** Changing to crops with a lower water demand, or to drip / trickle irrigation applied at night, which delivers water to the roots and loses far less to evaporation **1** Metering and pricing water, or setting abstraction licences and quotas, so that users have a reason to waste less **1** Recycling treated waste water for irrigation or industry, and harvesting rainwater, so less is taken from the aquifer **1** Lining canals and repairing leaking pipes so that less of the water abstracted is lost before it is used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **abstraction** → water taken out of a river, lake or aquifer • **Recycling** → reprocessing a material to make something new **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-121; AQA spec 3.2.2.1-3.2.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended **1** Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges **1** The particles clump together (flocs) and settle in the clarifier tank **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources **1** Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed – improves taste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) **1** Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**1 mark each** 1 mark for each of the 3 points below. **1 Ion exchange:** toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) **1** Reverse osmosis / distillation (desalination) - expensive, energy-intensive **1 Activated carbon:** organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients – worms create tunnels improving drainage and aeration **up to 2** Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds **up to 2** Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions – nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption **up to 2** Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ammonification** → ammonium released from dead organic matter and protein • **humus** → dark, decomposed organic matter in soil • **peds** → soil crumbs / aggregates of soil particles	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** This is the range of tolerance of most plants and soil biota – acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes **1** In alkaline soils phosphates become insoluble / unavailable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate **1** Infiltration falls so waterlogging is more likely and runoff / erosion increase **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Interstitial** → pore spaces / air spaces between the particles • **nitrification** → ammonium oxidised to nitrite and then nitrate • **Infiltration** → water soaking into the soil surface	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Asbestos:** fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma – disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) **2 Cyanide:** very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested – disposal: incineration so carbon and nitrogen are separated and oxidised to CO2 and NOx (much smaller problem) **2 High-level radioactive waste:** intense ionising radiation, heat generation, long half-lives – disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **vitrification** → sealing waste into glass **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
05.2	1 mark for the point below. **1** The fibres are bound / cannot become airborne so cannot be inhaled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material **1** The solid cement immobilises the waste so it cannot leak / leach out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
05.4	1 mark for the point below. **1** Even if the glass shatters the waste stays encapsulated in the fragments – glass is more stable / resistant to heat and leaching over very long timescales **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass • **leaching** → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **stratospheric ozone** → the ozone in the stratosphere that absorbs UV	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms **1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere **1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1 Negative feedback:** a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium **1 Positive feedback:** a change causes other changes that increase the original change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **Negative feedback** → a change that cancels itself out / a self-correcting cycle	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Increased low-level cloud:** higher temperatures increase evaporation and condensation, producing more cloud – clouds have a high albedo so more sunlight is reflected and warming is reduced **2 Increased photosynthesis:** higher temperatures and CO2 increase photosynthesis, removing CO2 – if stored in woody tissue atmospheric CO2 rises less **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
07.3	**1 mark each** 1 mark for each of the 4 points below. **1 Permafrost:** frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground **1** Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming **1 Methane hydrate:** methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature **1** If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **waterlogged** → soil pores filled with water so there is no air	4

Do not write outside the box

Question	Marking guidance	Marks
07.4	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Reduced albedo as snow / ice melts **1** Ocean acidification reducing coral survival and carbon sequestration **1** Increased forest and peat fires releasing CO2 **1** Faster decomposition of soil organic matter releasing CO2 **1** Increased water vapour (a greenhouse gas) in warmer air **1** Increased cirrus (high) cloud which traps infrared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
07.5	**1 mark each** 1 mark for each of the 2 points below. **1** A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable **1** So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
09.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** The readings are precise but may not be accurate **Also accept** • **precise** → readings close together / fine resolution • **accurate** → close to the true value	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** A result that differs from other results with which it was expected to be similar **1 Repeat the study / measurement:** if it recurs it is a 'real' result caused by another variable – if not it was an error and can be excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** To test the reliability of the method / equipment **1** To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Fossil fuel technologies:** secondary / tertiary recovery raising recovery from 20% to 60% – directional drilling and subsea wells – ROVs – fracking for tight oil and gas – tar sands / oil shales – coal gasification / liquefaction – enhanced gas recovery – methane hydrates – CCS (pre / post combustion, storage) • **Limits:** still finite (discoveries declining since the 1980s) – rising energy and financial cost of unconventional sources – environmental impacts (aquifers, water use, tremors, land take, habitat damage) – CO2 and other pollution – CCS unproven at scale and impractical for vehicles • **Nuclear:** new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash) – molten salt reactors – plutonium breeder and thorium reactors increasing fuel use many-fold – fusion as a long-term prospect • **Limits:** cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium – fusion not commercial • **Sustainability criteria:** maintaining supply without unacceptable environmental, economic or social impact – comparison with renewables and energy conservation – energy gap argument • **Judgement:** technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely	9

Do not write outside the box

Question	Marking guidance	Marks
	– role as transition / base-load alongside renewables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **fracking** → pumping fluid into rock at high pressure to release gas or oil • **CCS** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Technology has repeatedly extended fossil-fuel supplies. Secondary and tertiary recovery raise the proportion of oil recovered from about 20% to 60%; directional drilling, subsea wells and ROVs open deep-water fields; fracking releases tight oil and gas; tar sands and oil shales, coal gasification and liquefaction convert lower-grade resources; and carbon capture and storage could reduce the CO_2 impact. Nuclear fission can be stretched even further: uranium can be adsorbed from seawater with polymers or recovered from phosphate mining and coal ash, molten-salt reactors run more safely, and plutonium breeder and thorium reactors multiply the energy obtainable from each tonne of fuel, with fusion a long-term prospect.	
	None of this makes the resources renewable. Discoveries of conventional oil have declined since the 1980s, unconventional sources cost more energy and money per unit and damage aquifers, use huge volumes of water, cause tremors and take land, and CCS is unproven at scale. Nuclear projects overrun their budgets, decommissioning and long-lived waste remain unsolved, public opposition and proliferation constrain expansion, low-grade uranium has a high embodied energy, and fusion is not commercial.	
	Sustainability means maintaining supply without unacceptable environmental, economic or social impact. New technology can extend supplies for decades and cut some impacts, which matters for closing the energy gap while renewables and conservation scale up, but it cannot make non-renewable resources sustainable indefinitely. Their proper role is as a transition and base-load source alongside renewables, not a permanent solution.	
	Weaker answer - Level 2 (5 marks)	
	New technologies mean fossil fuels can last longer. Fracking gets gas out of shale rock and directional drilling reaches oil under the sea. Carbon capture and storage can stop the carbon dioxide going into the atmosphere. For nuclear power, breeder reactors and thorium reactors can use fuel more efficiently and fusion might work in the future. However fossil fuels are still non-renewable so they will run out eventually, and fracking can pollute groundwater. Nuclear waste is still a problem. So new technologies help but they cannot make fossil fuels and nuclear power fully sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 reaches a sensible conclusion but supports it thinly; Level 3 gives specific evidence (20% to 60% recovery, declining discoveries, embodied energy of low-grade uranium).	
	• Level 3 sets out a criterion for sustainability and judges against it - the hallmark of 'evaluate'.	
	• Level 3 places the technologies in context (transition and base load alongside renewables) rather than treating them in isolation.	
10.2	1 mark for the point below. **1** 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Negative feedback:** increased low-level cloud (albedo), increased photosynthesis and sequestration – hydrological cycle evaporation-precipitation • **Positive feedback:** melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • **Tipping points:** self-sustaining processes – why control must precede them – examples • **Implications:** urgency and precautionary principle – need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation – geoengineering risks • Difficulty of prediction because feedbacks interact at different scales – models – international agreements and per capita footprints • Judgement on whether current strategies are adequate	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

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